AI-Based Battery Lifecycle Assessment and Second-Life Management Using Digital Battery Passports

Week 1: Scoping, architecture and real data acquisition

Mon 2 Nov - Fri 6 Nov 2026 · Sprint 1 · Overall progress 10%

Student name		**Roll no.**	
Faculty in charge		**Date submitted**	Fri 6 Nov 2026

On track **This week in one line:** Fixed the project scope, chose two real battery-ageing datasets (NASA PCoE and CALCE CS2), designed the layered system architecture and set up the repository.

1. Goal for the week

Fix the problem statement and scope, find real battery data that covers the whole battery life, and design the system before writing models.

2. Day-by-day work

Day	Work done
Mon 2 Nov	Wrote the problem statement and five objectives: SOH estimation, RUL prediction, second-life grading, Digital Battery Passport, analytics interface.
Tue 3 Nov	Surveyed public Li-ion ageing datasets (NASA PCoE, CALCE, Oxford, MIT/Toyota) for size, signals and how far each cell is aged.
Wed 4 Nov	Chose NASA PCoE (impedance and temperature) plus CALCE CS2 (aged down to 15-27% SOH, so it covers second life). Downloaded both.
Thu 5 Nov	Studied the EU Battery Regulation 2023/1542: passport mandatory from 18 Feb 2027 (Art. 77, Annex XIII), access tiers, battery status values.
Fri 6 Nov	Designed the six-layer architecture, wrote config.py with all thresholds, created the repository and this report.

3. Completed deliverables

- **batterylife/config.py**: datasets, lifecycle thresholds (80% / 60% SOH), second-life application catalogue, sustainability assumptions.
- **scripts/download_data.py**: fetches the raw data.
- **docs/architecture.md** and the architecture diagram (below).

4. Results

Decision	Choice	Reason
Datasets	NASA PCoE (4 cells) + CALCE CS2 (4 cells)	Real data; together they cover first life, second life and recycling
First-life end	80% SOH	Common EV retirement threshold
Second-life floor	60% SOH	Below this re-use is not worthwhile
SOH inputs	Charging signals only	A battery management system sees them every charge
Passport	Signed, hash-chained events	Aligned with EU Regulation 2023/1542, Annex XIII

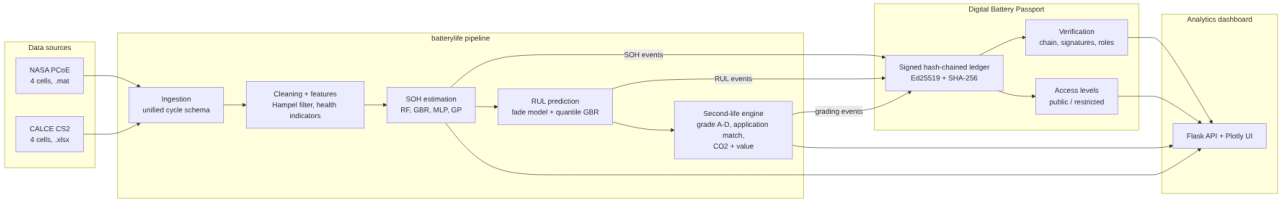


Figure 1. System architecture: data sources, pipeline, Digital Battery Passport and dashboard.

5. Problems faced and how they were solved

Problem	Resolution
Only 8 cells exist in total across both datasets.	Validate with leave-one-cell-out and report errors per cell.
CALCE raw data is about 170 MB.	Kept out of git; a download script fetches it.

6. Plan for next week

- Parse both datasets into one validated cycle-level table.
- Check data quality and document the findings.

Faculty remarks

Faculty signature: _____ Student signature: _____

AI-Based Battery Lifecycle Assessment and Second-Life Management Using Digital Battery Passports

Week 2: Data ingestion and quality checks

Mon 9 Nov - Fri 13 Nov 2026 · Sprint 2 · Overall progress 20%

Student name		**Roll no.**	
Faculty in charge		**Date submitted**	Fri 13 Nov 2026

On track **This week in one line:** Built loaders for NASA .mat and CALCE Excel files into one validated table: 4,631 cycles from 8 cells.

1. Goal for the week

Read both raw datasets into a single cycle-level table with one schema, and validate it.

2. Day-by-day work

Day	Work done
Mon 9 Nov	Defined the unified schema: 16 columns, one row per full charge-discharge cycle.
Tue 10 Nov	NASA loader: paired every discharge with its preceding charge; interpolated EIS resistance (Re + Rct) onto discharge times.
Wed 11 Nov	CALCE loader: read 100 Arbin Excel files in date order, renumbered cycles, kept only full 1C discharges.
Thu 12 Nov	Registry with CSV cache and validation; fixed two bugs found while checking (constant resistance, rest time counted as discharge).
Fri 13 Nov	Data-quality report, dataset document, raw-SOH figure, 5 unit tests, this report.

3. Completed deliverables

- **batterylife/data/**: schema.py, nasa.py, calce.py, registry.py.
- **data/processed/cycles_raw.csv** and data_quality.json.
- **docs/dataset.md**; tests/test_data.py (5 tests).

4. Results

Dataset	Cells	Cycles	SOH range	Extra signals
NASA PCoE	B0005, B0006, B0007, B0018	636	102% to 58%	Impedance, temperature
CALCE CS2	CS2_35 to CS2_38	3,995	104% to 15%	Internal resistance
Total	8	4,631		

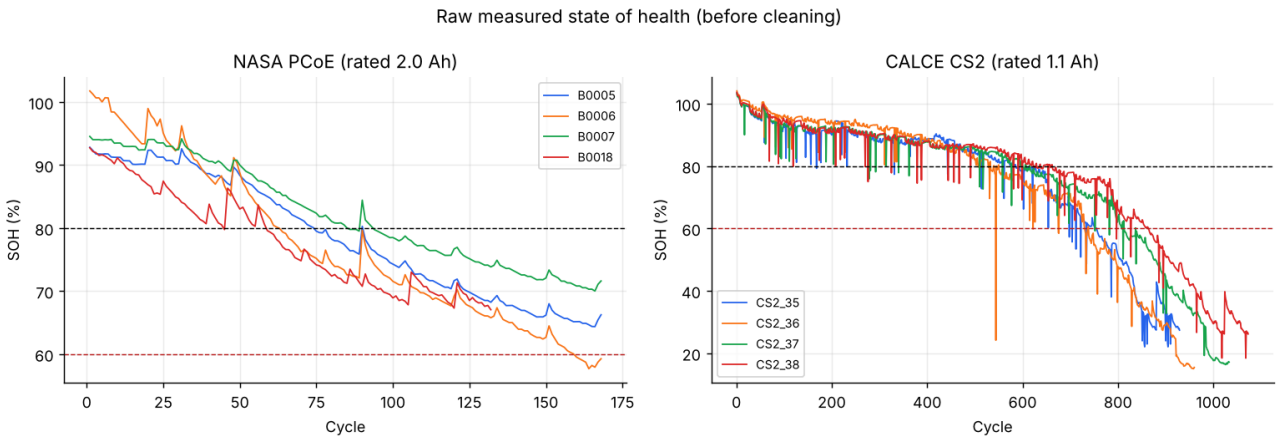


Figure 2. Measured state of health of all 8 cells before cleaning. Dashed lines: 80% and 60%.

5. Problems faced and how they were solved

Problem	Resolution
NASA resistance came out constant for every cycle.	pandas 3 stores dates in microseconds; switched to Python datetime timestamps.
NASA discharge time included rest after cut-off.	Kept only the period with load current above 1 A.
Interrupted CALCE cycles appear as sudden drops.	Planned for next week's cleaning step.

6. Plan for next week

- Remove interrupted cycles with a robust filter.
- Build health indicators from the charging curve.

Faculty remarks

Faculty signature: _____ Student signature: _____

AI-Based Battery Lifecycle Assessment and Second-Life Management Using Digital Battery Passports

Week 3: Data cleaning and health indicators

Mon 16 Nov - Fri 20 Nov 2026 · Sprint 3 · Overall progress 30%

Student name		**Roll no.**	
Faculty in charge		**Date submitted**	Fri 20 Nov 2026

On track **This week in one line:** A one-sided Hampel filter removed 120 interrupted cycles while keeping real capacity regeneration; charge-curve health indicators correlate up to r = 1.00 with SOH.

1. Goal for the week

Remove measurement artefacts and build health indicators that a battery management system can observe during ordinary charging.

2. Day-by-day work

Day	Work done
Mon 16 Nov	Analysed the outliers: interrupted discharges (drops) versus capacity regeneration after rest (real, jumps up).
Tue 17 Nov	Implemented a one-sided Hampel filter (11-cycle window, 3 x MAD, two passes) that removes only drops.
Wed 18 Nov	Two-sided spike repair for charge features; 9-cycle median SOH for end-of-life detection.
Thu 19 Nov	Built health indicators: CC charge time, CV charge time, 3.9-4.2 V rise time, resistance, cycle count; per-cell relative versions.
Fri 20 Nov	Correlation analysis, figure, 7 new tests (12 total), this report.

3. Completed deliverables

- **batterylife/features/cleaning.py** and **indicators.py**.
- **data/processed/features.csv**.
- tests/test_features.py (7 tests).

4. Results

Health indicator	Correlation with SOH, NASA	Correlation with SOH, CALCE
CC charge time	0.94	1.00
3.9-4.2 V rise time	0.93	0.97
CV charge time	-0.90	-0.73
Resistance	-0.56	-0.77
Outliers removed	0 of 636 (regeneration kept)	120 of 3,995

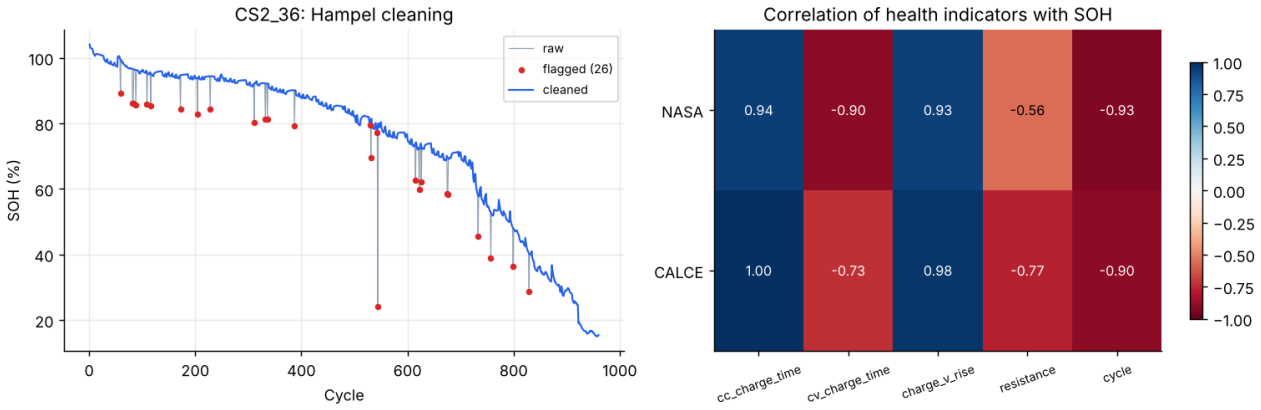


Figure 3. Left: cleaning on CS2_36 (red points removed). Right: correlation of each indicator with SOH.

5. Problems faced and how they were solved

Problem	Resolution
CALCE cells charge from empty, so CC charge time is almost a direct capacity measurement.	Added a 'partial-charge' feature set without it, tested next week.
CS2_38 resistance sensor shows a long plateau (cycles 30-140).	Documented as a sensor artefact; robust baseline limits its effect.

6. Plan for next week

- Train and compare SOH models with leave-one-cell-out validation.
- Add calibrated confidence intervals.

Faculty remarks

Faculty signature: _____ Student signature: _____

AI-Based Battery Lifecycle Assessment and Second-Life Management Using Digital Battery Passports

Week 4: State-of-health estimation models

Mon 23 Nov - Fri 27 Nov 2026 · Sprint 4 · Overall progress 42%

Student name		**Roll no.**	
Faculty in charge		**Date submitted**	Fri 27 Nov 2026

On track **This week in one line:** Five models compared on cells never seen in training. SOH error 2.20% (NASA, Gradient Boosting) and 0.81% (CALCE, Gaussian Process), with 95% intervals covering 95-98% of true values.

1. Goal for the week

Estimate SOH from charging signals only, validated honestly on unseen cells, with calibrated uncertainty.

2. Day-by-day work

Day	Work done
Mon 23 Nov	Leave-one-cell-out harness: each cell is scored by a model trained only on the other three cells of its dataset.
Tue 24 Nov	Trained Ridge, Random Forest, Gradient Boosting, MLP and Gaussian Process; compared relative vs absolute features.
Wed 25 Nov	Switched to absolute features (NASA error fell from 4.1% to 2.2%); added the partial-charge feature set.
Thu 26 Nov	GP intervals were over-confident (66% coverage); replaced with split-conformal 95% intervals.
Fri 27 Nov	Saved production models, figure, 4 new tests (16 total), this report.

3. Completed deliverables

- **batterylife/models/soh.py** and evaluation.py.
- **results/soh_metrics.json**, soh_predictions.csv.
- tests/test_soh.py.

4. Results

Model	NASA full	NASA partial	CALCE full	CALCE partial
Ridge	2.35	2.96	0.83	5.87
Random Forest	2.41	2.49	1.15	1.96
Gradient Boosting	2.20	2.49	1.32	2.07
MLP	5.01	7.07	7.40	3.28
Gaussian Process	3.82	4.20	0.81	3.83

RMSE in SOH percentage points on unseen cells. 'Partial' leaves out CC charge time (charge may start part-way).

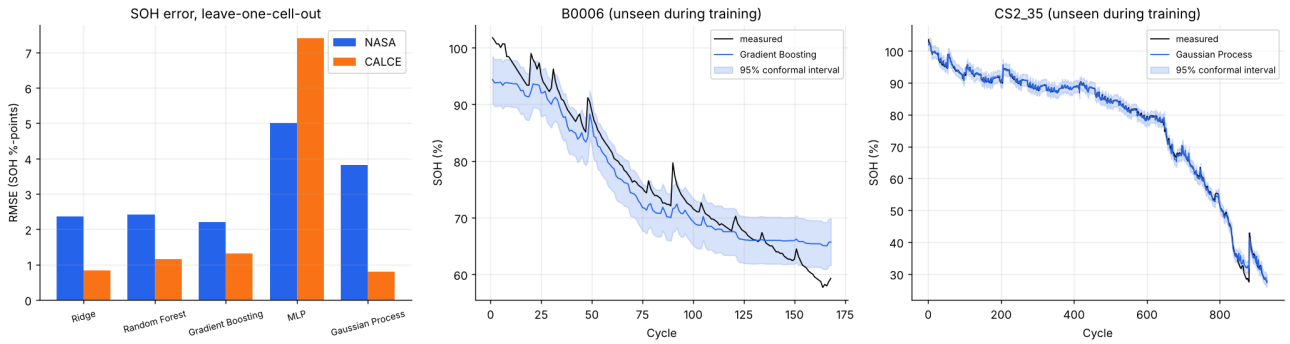


Figure 4. Left: error by model. Middle and right: estimate and 95% interval on two unseen cells.

5. Problems faced and how they were solved

Problem	Resolution
Relative features erased differences in starting capacity between NASA cells.	Absolute features adopted.
Tree models cannot extrapolate below the training range (B0006 at 58%).	Documented; more deep-fade cells would fix it.
MLP underperforms with only 3 training cells.	Kept only as a baseline.

6. Plan for next week

- Predict remaining useful life to 80% and 60% SOH.
- Calibrate the uncertainty bands.

Faculty remarks

Faculty signature: _____ Student signature: _____

AI-Based Battery Lifecycle Assessment and Second-Life Management Using Digital Battery Passports

Week 5: Remaining useful life prediction

Mon 30 Nov - Fri 4 Dec 2026 · Sprint 5 · Overall progress 55%

Student name		**Roll no.**	
Faculty in charge		**Date submitted**	Fri 4 Dec 2026

On track **This week in one line:** Quantile Gradient Boosting predicts cycles left to 80% SOH within 8.2 cycles (NASA) and 46.6 cycles (CALCE); calibrated bands now cover 76-84% at an 80% target.

1. Goal for the week

Predict how many cycles remain before 80% SOH (end of first life) and 60% SOH (end of second life), with calibrated uncertainty.

2. Day-by-day work

Day	Work done
Mon 30 Nov	Ground-truth end-of-life from smoothed SOH; causal snapshots so the predictor only sees cycles 1..k.
Tue 1 Dec	Baselines: linear trend and an empirical fade model with an accelerating 'knee' term.
Wed 2 Dec	Quantile Gradient Boosting (10/50/90%) on SOH estimate, fade rate, fade acceleration, charge signals, resistance.
Thu 3 Dec	Raw bands covered only 22-33%; implemented conformalised quantile regression (CQR).
Fri 4 Dec	Evaluation by life stage, figure, 5 new tests (21 total), this report.

3. Completed deliverables

- **batterylife/models/rul.py**.
- **results/rul_metrics.json**, rul_predictions.csv.
- tests/test_rul.py.

4. Results

Dataset	Mean life to 80%	Linear trend	Fade model	Quantile GBR	Band coverage
NASA	72 cycles	78.9	55.9	**8.2**	75.9%
CALCE	578 cycles	388.8	145.4	**46.6**	78.6%
CALCE to 60%	790 cycles	499.4	352.8	**55.4**	84.0%

Mean absolute error in cycles on unseen cells.

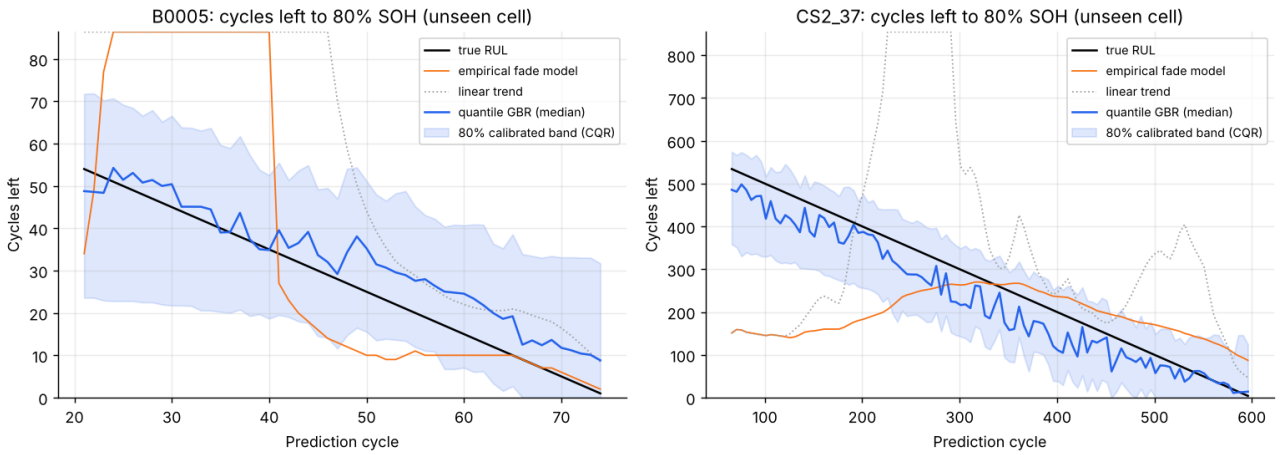


Figure 5. Cycles left to 80% SOH on two unseen cells: truth, baselines and the calibrated band.

5. Problems faced and how they were solved

Problem	Resolution
Curve extrapolation cannot see the CALCE knee before it happens.	The learned model anticipates it from the other cells.
NASA capacity regeneration flattens fitted trends.	Same: data-driven model; baselines kept for comparison.
Uncalibrated bands were dangerously narrow.	CQR brings coverage close to the 80% target.

6. Plan for next week

- Turn SOH and RUL into second-life grades and application matches.
- Quantify CO2 and value of re-use.

Faculty remarks

Faculty signature: _____ Student signature: _____

AI-Based Battery Lifecycle Assessment and Second-Life Management Using Digital Battery Passports

Week 6: Second-life grading, application matching and value

Mon 7 Dec - Fri 11 Dec 2026 · Sprint 6 · Overall progress 66%

Student name		**Roll no.**	
Faculty in charge		**Date submitted**	Fri 11 Dec 2026

On track **This week in one line:** Uncertainty-aware grader agrees with lab-truth grades in 83.2% of 101 decisions (89.5% when confident) and flags 65% of its own mistakes for a lab test.

1. Goal for the week

Decide what each retired battery should do next, with honest confidence, and quantify the value of re-use.

2. Day-by-day work

Day	Work done
Mon 7 Dec	Application catalogue: A (EV re-use), B (grid/home storage), C (backup power), D (recycle) with SOH, resistance and cycle limits.
Tue 8 Dec	Rule engine with reasons; confidence check by re-grading at the lower 95% bound of SOH.
Wed 9 Dec	Service years, CO2 avoided and value, scaled to a 60 kWh pack.
Thu 10 Dec	Evaluated at 15 retirement levels per cell against lab-truth grades; confusion matrix.
Fri 11 Dec	Figure, 12 new tests (33 total), this report.

3. Completed deliverables

- **batterylife/lifecycle/**: grading.py, sustainability.py, second_life.py.
- **results/second_life_assessments.csv** and summary.
- tests/test_lifecycle.py.

4. Results

Metric	Result
Grade agreement with lab truth	83.2% (NASA 87.8%, CALCE 80.0%)
Agreement when confidence is high	89.5%
Mistakes already flagged low-confidence	64.7%
Conservative / optimistic errors	7 / 10
Second-life length error (CALCE)	33.6 cycles
Per 60 kWh pack at 80% retirement	about 3.6 t CO2e avoided (upper bound), USD 2,478 re-use value

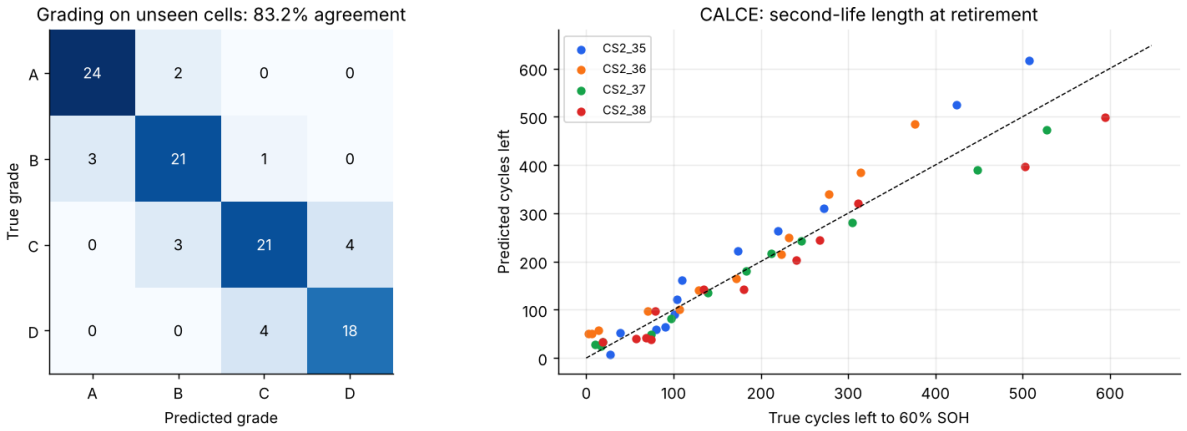


Figure 6. Left: AI grade vs lab grade on unseen cells. Right: predicted vs true second-life length.

5. Problems faced and how they were solved

Problem	Resolution
Most wrong grades sit right at a grade boundary.	The low-confidence flag catches them and asks for a lab test.
Only one NASA cell reaches 60% SOH.	Cycle criterion applied only where ground truth exists (CALCE).
Lab ageing is harsh, so service years look short.	Reported as a relative ranking, not a forecast.

6. Plan for next week

- Build the signed, hash-chained Digital Battery Passport.
- Test it against tampering attacks.

Faculty remarks

Faculty signature: _____ Student signature: _____

AI-Based Battery Lifecycle Assessment and Second-Life Management Using Digital Battery Passports

Week 7: Digital Battery Passport

Mon 14 Dec - Fri 18 Dec 2026 · Sprint 7 · Overall progress 78%

Student name		**Roll no.**	
Faculty in charge		**Date submitted**	Fri 18 Dec 2026

On track **This week in one line:** Built a tamper-evident passport: every event is SHA-256 chained and Ed25519 signed. 8 passports with 164 events verify; all 5 tested attacks are detected.

1. Goal for the week

Record each battery's whole life in a passport that anyone can verify and no one can secretly change, aligned with EU Regulation 2023/1542.

2. Day-by-day work

Day	Work done
Mon 14 Dec	Schema: 10 event types, 6 actor roles (manufacturer, OEM, BMS, assessor, repurposer, recycler), required fields, public and restricted access.
Tue 15 Dec	Ed25519 key pairs per actor; canonical JSON hashing; signed, hash-chained ledger.
Wed 16 Dec	Verification of chain links, content hashes, signatures, role authorisation, required fields and chronology.
Thu 17 Dec	Public (QR-code) view hiding restricted health data; attack tests. Fixed a bug where signed data could be changed by the caller.
Fri 18 Dec	Issued passports for all 8 cells, added the live tamper demo to the website, this report.

3. Completed deliverables

- **batterylife/passport/**: schema.py, crypto.py, ledger.py, store.py, issue.py.
- **results/passports/**: 8 full and 8 public passports.
- tests/test_passport.py (10 tests).

4. Results

Attack on a passport	Detected by
Raise a past SOH value	Content hash
Raise a value and recompute every hash	Ed25519 signature
Delete an event	Chain link
Back-date an event	Chronology check
Re-sign with another actor's key	Signature

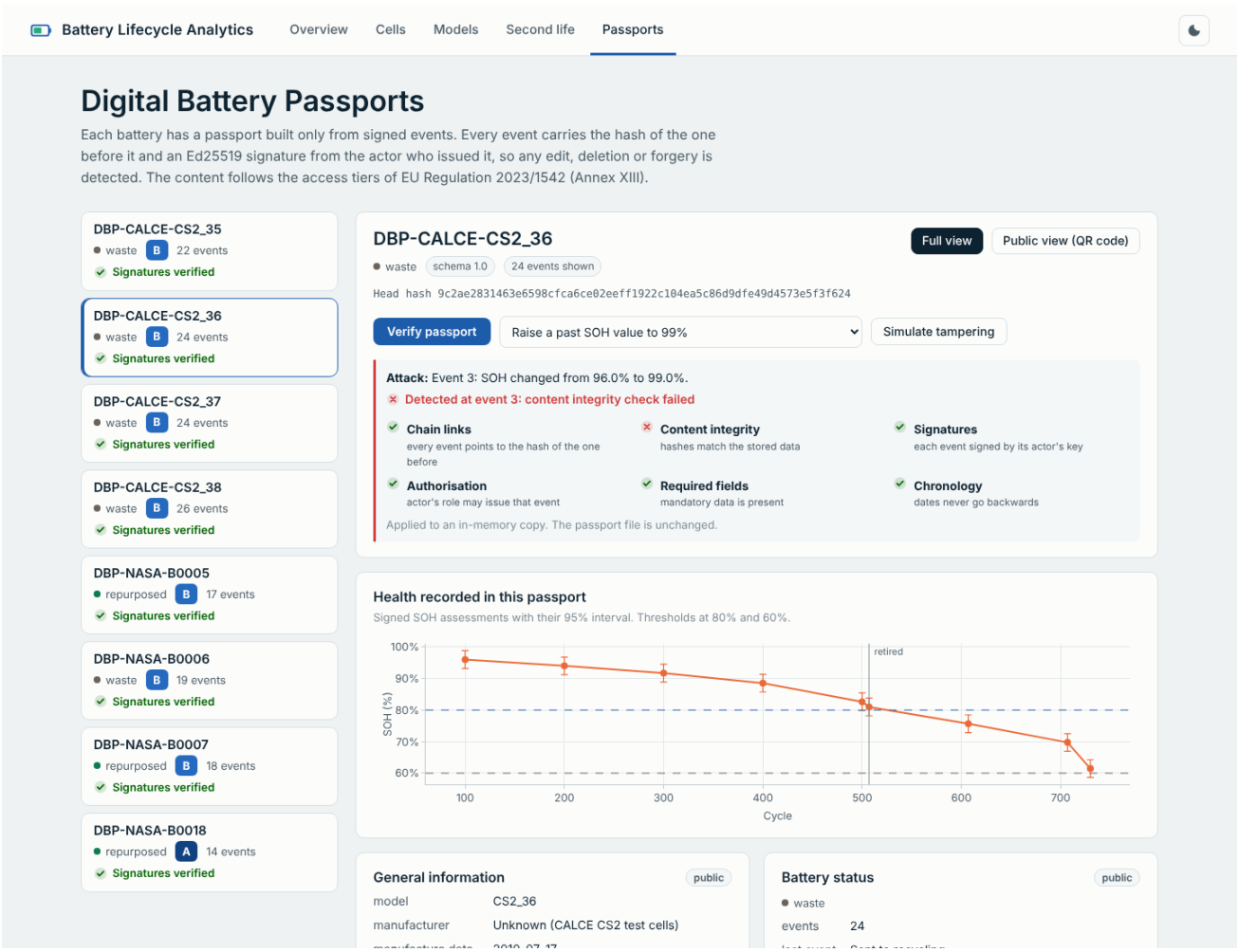


Figure 7. Website passport view: a forged SOH value is caught at event 3 and highlighted in the ledger.

5. Problems faced and how they were solved

Problem	Resolution
Signed event data was shared with the caller's object.	Data is deep-copied at signing time; a test guards it.
Timestamps went backwards after repurposing.	Administrative events now follow by hours; chronology is a verification check.
A single-store ledger cannot stop deletion of a whole file.	Future work: anchor each chain head in a public registry.

6. Plan for next week

- Integrate all stages into one pipeline and command-line tool.
- Pipeline tests and a full reproducibility run.

Faculty remarks

Faculty signature: _____ Student signature: _____